Muhammad Shoaib

Data Science Student | Machine Learning & Computer Vision
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PROFESSIONAL SUMMARY

Final-year Data Science student at CUSIT, Peshawar. Most of my work is in machine learning and computer vision, and lately I've been building with LLMs too. My two main projects are Jikimi AI, a CCTV system that warns guards about street crime in Urdu, and ParkinSense PK, which tracks how Parkinson's patients move between clinic visits. I'm looking for an AI/ML internship or junior role.

EDUCATION

City University of Science and Information Technology (CUSIT) — Peshawar, Pakistan
BS in Data Science | CGPA: 3.18 / 4.00 — 2023 – 2027 (Expected)

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, Java
Machine learning & deep learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers, NLP, feature engineering, fine-tuning, federated learning, statistical inference
Computer vision: OpenCV, MediaPipe, YOLO, object tracking, pose estimation, ONNX, OpenVINO
Generative AI: LLMs, LangChain, LangGraph, LangSmith, RAG, multi-agent systems, prompt engineering, OpenAI API
Backend & MLOps: FastAPI, Flask, Streamlit, REST APIs, Docker, Git, GitHub Actions, DVC
Databases: PostgreSQL, ChromaDB, Pinecone
Data analysis: Pandas, NumPy, SciPy, Matplotlib, Seaborn, Plotly, librosa

EXPERIENCE

Automation and Deployment Engineer | MA Soft Tech

PROJECTS

Jikimi AI | street-crime alerts for CCTV cameras — Oct 2026
github.com/shiffu-ds/Jikimi-Ai

- Built a system that runs on a laptop beside the CCTV cameras shops and societies already have. It warns guards about bike snatching and notices bikes circling the same place before anything happens. No face recognition.
- Added personal-safety alerts based only on behaviour, such as a phone snatched on foot, someone being followed or surrounded, a fall, a bike being stolen or a crowd running. It never guesses gender, age or identity.
- Trained a pose-based ST-GCN snatch detector in PyTorch on about 9,000 synthetic samples and got 0.978 test AUC. Quantizing the YOLO models to INT8 with OpenVINO let an old i5 laptop handle 1.3 cameras instead of 0.9.
- Tried federated learning on three simulated sites. It got 0.944 AUC against 0.954 for pooled data, and no pose data left a site. Differential privacy at that size either broke the model or protected nothing.
- Alerts come with a short reason in Urdu, and guards can search clips in Roman Urdu, like "kal raat kaali bike 2 sawar". In 30 hours of simulated normal traffic, it gave zero false alarms.
- Sends alerts by SMS during load-shedding, with a FastAPI and PostgreSQL backend and an Urdu dashboard. It has 381 automated tests, but so far it has only run on a simulator and one real traffic video.

ParkinSense PK | movement tracking for Parkinson's patients — Oct 2026
github.com/shiffu-ds/Parkinsense_PK

- A web and Android app where Parkinson's patients record short tasks like walking, finger tapping, spiral drawing and voice. Their clinician sees how each result changes between visits.
- Movement is measured from video with MediaPipe and OpenCV, and voice with librosa. Each recording passes a quality check before the patient gets a 3D replay, simple statistics and a printable report.
- On public walking videos it accepted a clean clip and asked for a retake when tracking dropped for 1.1 seconds. That test also caught a bug where arm swing came from a barely visible wrist, which I fixed.
- Runs on PostgreSQL and FastAPI with separate patient, clinician and admin access, in English, Urdu and Roman Urdu. It has 536 automated tests, but it doesn't diagnose anything and isn't clinically validated yet.

CERTIFICATIONS

The Ultimate Job Ready Data Science Course, CodeWithHarry — Sep 2025

ACHIEVEMENTS

Top 20 at atomcamp's National AI Hackathon (Agentic AI) with our team project, Burner AI — May 2026